

# NANOCHAT-FILIPINO P1.1 - WikiText-TL-39 fixed-budget depth vs held-out BPB (#306780)

**Author(s)**

This pre-registration is currently anonymous to enable blind peer-review.  
It has one author.

**Pre-registered on:**  
2026/08/15 19:34 (PT)

**1) Have any data been collected for this study already?**

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

**2) What's the main question being asked or hypothesis being tested in this study?**

Does increasing nanochat depth from 8 to 20 reduce held-out Tagalog bits-per-byte (BPB) when every depth sees the same model-visible token budget  $D_{3x} = 3 * T_{train}$  on WikiText-TL-39? Prediction: val BPB falls then flattens, or the train-val gap widens. We will not claim deeper is always better. This is not a CORE, chat, or classification test.

**3) Describe the key dependent variable(s) specifying how they will be measured.**

Primary DV: val\_bpb\_full on the held-out val split. BPB = mean token NLL /  $(\ln(2) * \text{mean UTF-8 bytes per token})$ , nanochat token\_bytes.pt, special tokens excluded; full val set after training (not --eval-tokens). Secondary: test\_bpb, same formula/packing/T=2048, once, after freezing the best-val checkpoint. Selection uses val only.

**4) How many and which conditions will participants be assigned to?**

No human participants. Four confirmatory depths (8, 12, 16, 20), same WikiText-TL-39 package, same train-only 32768 BPE, same 70/15/15 split, T=2048, budget  $D_{3x} = 3 * T_{train}$  (actual tokens = num\_iterations \* total\_batch\_size). Native --target-param-data-ratio=12 runs, if any, are not equal-exposure comparisons.

**5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.**

Compare val\_bpb\_full across the four  $D_{3x}$  depths. Also report the train-val BPB gap. Each trained val\_bpb\_full must be below (i) an untrained same-depth model and (ii) a train-fitted UTF-8 byte unigram on the same val bytes.  $D^*$  = lowest val\_bpb\_full; test\_bpb is for  $D^*$  and is not used to choose  $D^*$ . One seed: point estimates; ignore gaps < 0.01 BPB. Three seeds: mean and sample SD. No token-level CIs. No CORE in the confirmatory comparison.

**6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.**

Drop a unit only if null/empty after LF-normalize, or length > 200000 characters. If drops exceed 5%, stop. Split is invalid if any exact text hash is in more than one of train/val/test. Exclude a run from the confirmatory table only for NaN/Inf BPB, non-WikiText-TL-39 data in the train dir, test file in the train dir, val/test used in tokenizer training, target-param-data-ratio=-1, or a silent change of depth/T after OOM. Do not drop a depth because BPB looks bad. d4 smoke (20-100 steps) is not confirmatory.

**7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.**

Corpus is fixed: linkanjarad/Wikitext-TL39 data/train.parquet. No added documents. Use original 2019 splits if found; else sort sha256(utf-8 text) and take 70/15/15.  $T_{train}$  = BPE tokens in train after tokenizer training. Each confirmatory run:  $D_{3x}$  tokens, four depths, one seed. Extra seeds at 8 and 12 only if resources allow, and before looking at those seeds' test BPB.

**8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)**

Q1: Cruz and Cheng (2019) already collected the public corpus. We have not trained the confirmatory models or computed val\_bpb/test\_bpb. This files the analysis plan before those outcomes exist. Secondary (not used to pick  $D^*$  or the tokenizer): Tagalog vs GPT-2 bytes/token; native-ratio vs  $D_{3x}$  corpus passes; d8/d12  $D_{1x}$  pilots. Exploratory: clean ablation, Moses detok, other vocabs, d24,  $D_{10x}$ , CORE, downstream tasks. Canonical text is the source (no detokenize). Test data stay outside the train directory.